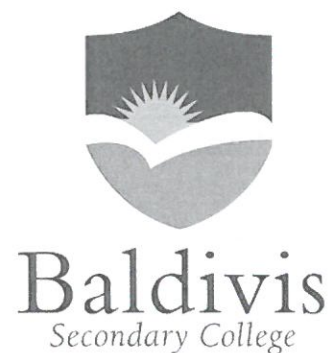


Name: Solutions.

Teacher: _____

Class: _____



**YEAR 8 EXAM
SEMESTER ONE 2015**

Section Two: Resource Allowed

TIME ALLOWED FOR THIS PAPER

Reading time before commencing 3 minutes.

Working time for paper 32 minutes.

MATERIAL(S) REQUIRED / RECOMMENDED FOR THIS PAPER

TO BE PROVIDED BY THE SUPERVISOR:

This Question / Answer Booklet

TO BE PROVIDED BY THE CANDIDATE:

Standard items: pens, pencils, pencil sharpener, eraser, correction fluid, ruler, highlighters.

Special items: Calculator, 1 x A4 sheet of notes.

Important note to candidates:

No other items may be used in this section of the examination. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Instructions to candidates

- Candidates are to remain seated during the exam and are not allowed to talk to other candidates.
- Mobile phones are not to be used at all during the exam.
- Students who fail to follow examination rules will be removed and dealt with in accordance to the schools policy.

Write your answers in the spaces provided in this Question/Answer Booklet. Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

- Planning: If you use spare pages for planning, indicate this clearly at the top of the page
- Continuing an answer: If you need the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

Show all your working clearly. Your working should be sufficient to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat

<i>Question</i>	<i>Out of</i>	<i>Mark</i>
1	5	
2	2	
3	2	
4	2	
5	6	
6	3	
7	4	
8	5	
9	6	
TOTAL	35	
Your Mark	/35	
Percentage		

Question 1

[5 marks]

Match the situation to its probability

A woman is pregnant with her 4 th baby and already has 3 sons what is the probability of her having a daughter		1/6	✓
The probability if you flipped a coin twice that you would get 2 tails		1/8	✓
If you place all the months of the year into a bag, what is the probability you draw a month starting with 'A' out		1/2	✓
If you place the letters BALDIVIS in a bag what is the probability of drawing a B		1/3	✓
In a bag you place 3 red, 2 green and 4 blue marbles. What is the probability of choosing a red marble?		1/4	✓

Question 2

[2 marks]

Use the order of operations to evaluate the following:

(a) $\frac{45 \times 2}{5+10}$ $\frac{90}{15} = 6$ ✓

(b) $\frac{-5}{1-3 \times 2}$ $\frac{-5}{-5} = 1$ ✓

Question 3

[2 marks]

In the expression $5y + 7x + 6$

(a) What is the coefficient of x? 7 ✓

(b) What is the constant term? +6 ✓

Question 4**[2 marks]**

Philip thought of a number (n), he then doubles it and adds 15.

- (a) Write an expression for this situation

$$2n + 15$$



- (b) Find the value of this expression when n is equal to 13.

$$2(13) + 15 = 41$$

**Question 5****[6 marks]**

Simplify the following:

- (a) $2a + 7a$

$$9a$$



- (b) $3x - 4y - x + 6y$

$$2x + 2y$$



- (c) $6m \times 7n$

$$42mn$$



(d) $\frac{9a^2b}{3ac}$

$$\frac{3ab}{c}$$



- (e) $5(a + 2) + 3a$

$$5a + 10 + 3a = 8a + 10$$



- (f) $a(4b + 8c) + 2a(3b - 3c)$

$$4ab + 8ac + 6ab - 6ac$$

$$= 10ab + 2ac$$



Question 6**[3 marks]**

Expand:

(a) $3(4 + x)$

$12 + 3x$



(b) $-5(y - 7)$

$-5y + 35$



(c) $3x(4y - 10x + 3)$

$12xy - 30x^2 + 9x$

**Question 7****[4 marks]**

Factorise the following expressions:

(a) $7a + 21$

$7(a + 3)$



(b) $2a + 4b - 10$

$2(a + 2b - 5)$



(c) $4x^2y + 2xy^3 + 8x^2$

$2x(2xy + y^3 + 4x)$



Question 8**[5 marks]**

Grace had completed 10 tests in Science in Year 8 and constructed the stem and leaf diagram below of her scores out of 100.

Key: $3|7 = 37$

Stem	Leaf
1	2
2	
3	7
4	
5	3 7
6	2 3 6 8
7	5 7

- a) Calculate Grace's mean and median for her Science Tests

mean: $\frac{570}{10} = 57$



median: 62.5



- b) Grace had been on an extended holiday in Bali with her family and missed nearly an entire topic so she scored only 12%. If she removed this score from her calculation how would the mean and median be affected?

mean: $\frac{558}{9} = 62$



median: 63.

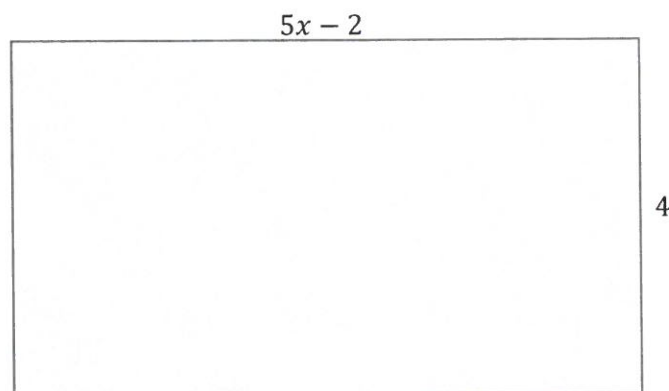


Both would increase as the lowest score had been removed.



Question 9**[6 marks]**

Andrew has a paddock on his new farm but has only been given the information below to help him:



- a) Write an expression for the perimeter of the paddock and simplify where possible

$$\begin{aligned} & 2(5x - 2) + 2(4) \\ &= 10x - 4 + 8 \\ &= 10x + 4 \end{aligned}$$



- b) Using your expression from (a) find the perimeter when $x = 8$

$$\begin{aligned} & 10(8) + 4 \\ &= 80 + 4 \\ &= 84 \end{aligned}$$



- c) Write an expression for the area of the paddock in expanded form

$$\begin{aligned} & 4(5x - 2) \\ & 20x - 8 \end{aligned}$$



